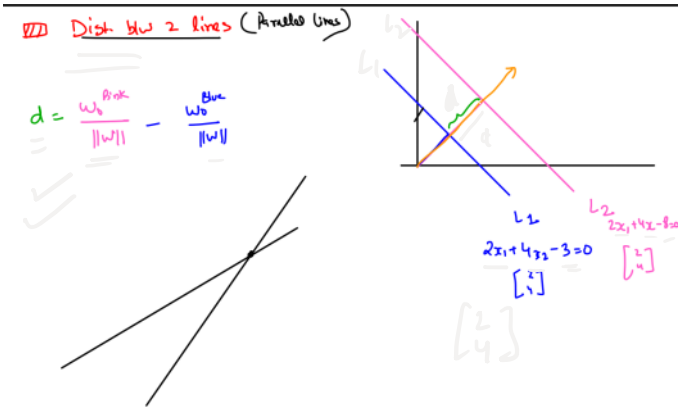
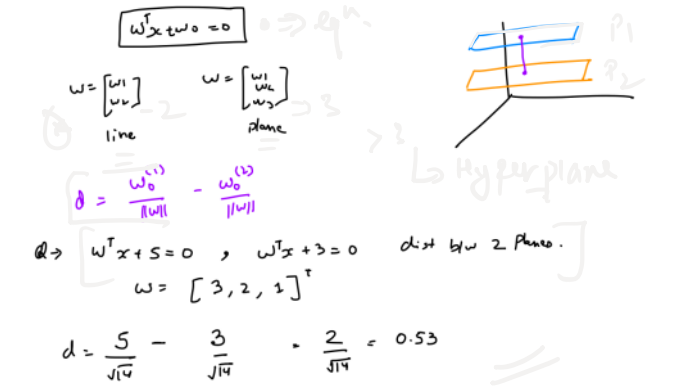
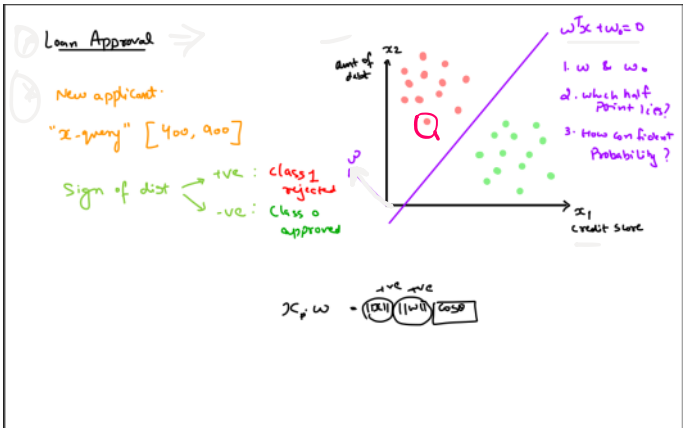
(*) PERCEPTRON ⇒

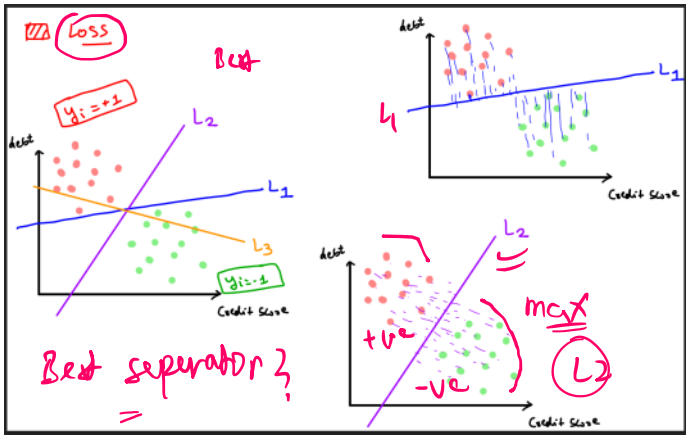
Agenda ⇒ (1) Recap
(2) VECTOR ADDITION
(3) PERCEPTRON ⇒ 1st ML algorithm
(4) Code ⇒ PERCEPTRON
(5) PROBLEMS (if time permits)

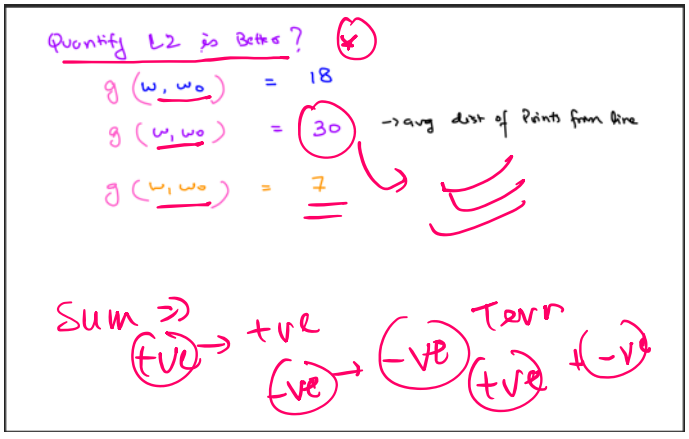


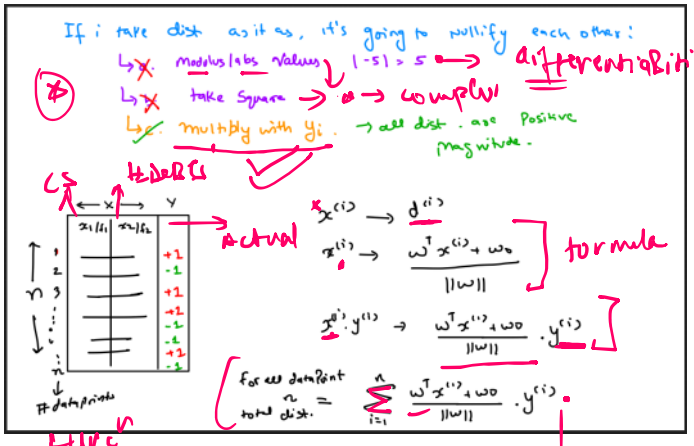




→ Best $w, \& w_0$
→
→







differentiability

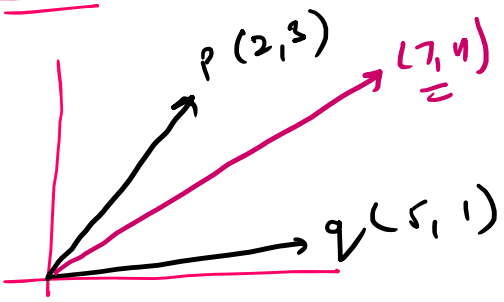
+ve , +ve = +ve

-ve , -ve = +ve

Total distance of all the points

⇒ avg.

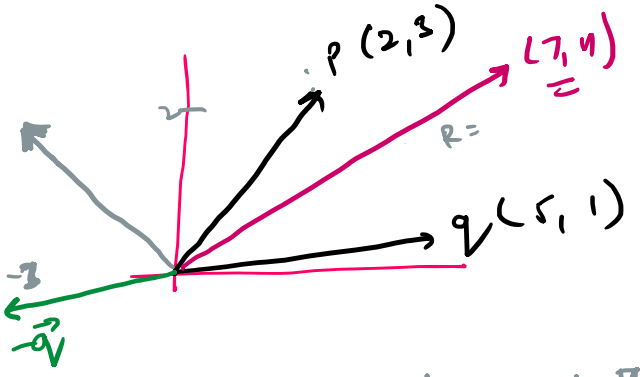
⇒ VECTOR ADDITION ⇒ $\vec{R} = \vec{P} + \vec{Q} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$



⇒ add 2 vectors
↓
Resultant → B/W

Subtraction of these vectors ⇒

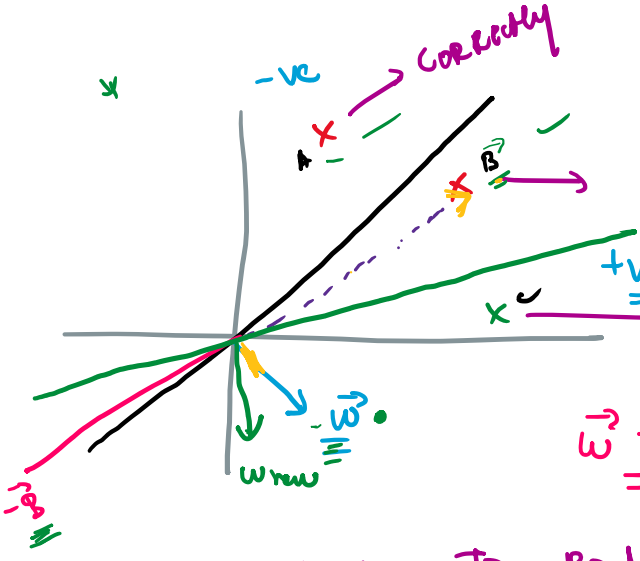
$\vec{P} - \vec{Q}$ ⇒ $\vec{P} + (-\vec{Q})$



$\begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} -5 \\ -1 \end{bmatrix}$

$= \begin{bmatrix} -3 \\ 2 \end{bmatrix}$

⇒ addⁿ of 2 vectors ⇒ resultant lies in B/W



good classifier? X

misclassified ⇒ to be on other side

new line

$\vec{w} + b$ ⊗ X

$\vec{w} + (-\vec{B})$

⇒ opportunity to Rotate the line ⇒ anticlockwise or clockwise

Rotate the w

50%

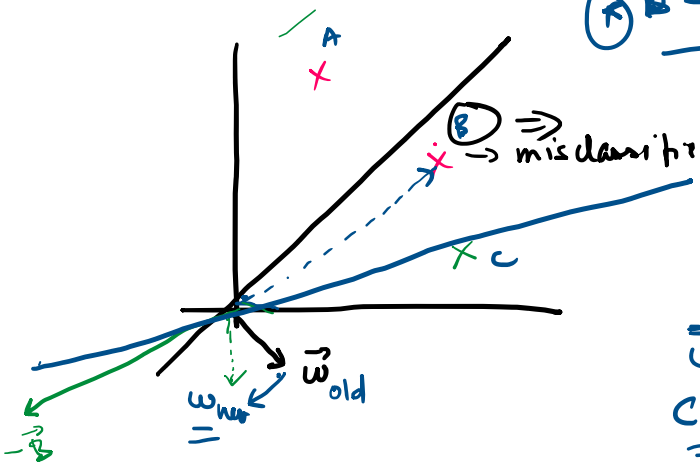
50%

green → +ve
red → -ve

Rotate the w ⇒ clockwise

ALGORITHM ⇒ $\vec{w}$, w_0

(*) $\vec{B}$ → -ve class



$\vec{w} + \vec{B}$ ✗ ✓

$\vec{w} - \vec{B}$ ⇒ ✓

clockwise ⇒ ✓

→ PERCEPTRON → Randomly

(*) PERCEPTRON ✗

(*) misclassification for -ve point

⇒ $w_{new} = w_{old} - \vec{B}$ (*) ⇒ clockwise dirn

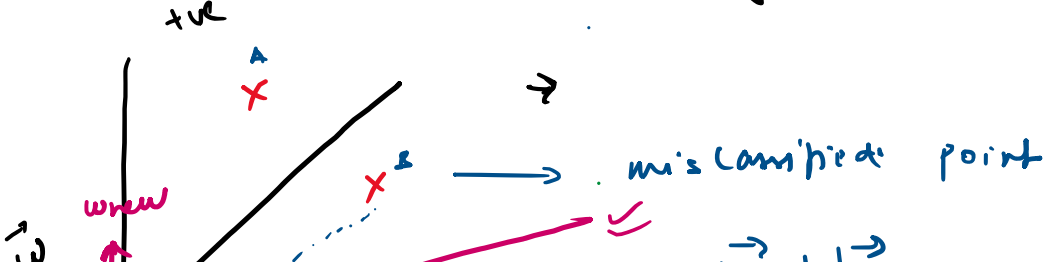


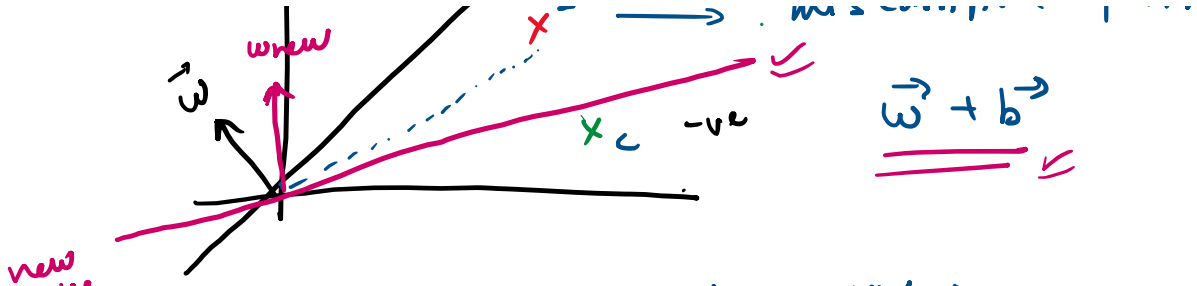
$\neq y_i$ ⇒ misclassified

⇒ distance → +ve / -ve

Case 2 ⇒

Red → +ve
green → -ve





$\vec{w} + b\vec{}$

⇒ Rotate w in such a way ⇒

+ve ⇒ $\vec{w} + b\vec{}$

update Rule of w when misclassification happend in +ve class

$\vec{w}$ ⇒ w_1, w_2 ⇒ avg distance max.

w_0 → shifting

w_0 ⇒ w_0 $\vec{w}$ X

UPDATE Rule ⇒

(1) if -ve point x^i is misclassified

⇒ $w - \alpha x_i$

(2) if +ve point x_i is misclassified

⇒ $w + x_i$

⇒ if any point with label y_i is misclassified

update rule ⇒ $w_{new} = w_{old} + x_i \times y_i$

Actuals ← y_i = +ve ⇒ $w = w + x_i$

y_i = -ve ⇒ $w = w - x_i$

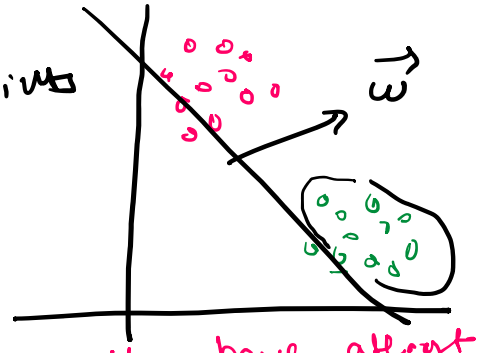
① PERCEPTRON ⇒

② goal → find good line ⇒

$[w_1, w_0]$

① step ⇒ initialize random w_1 w_0 ⇒ Random line

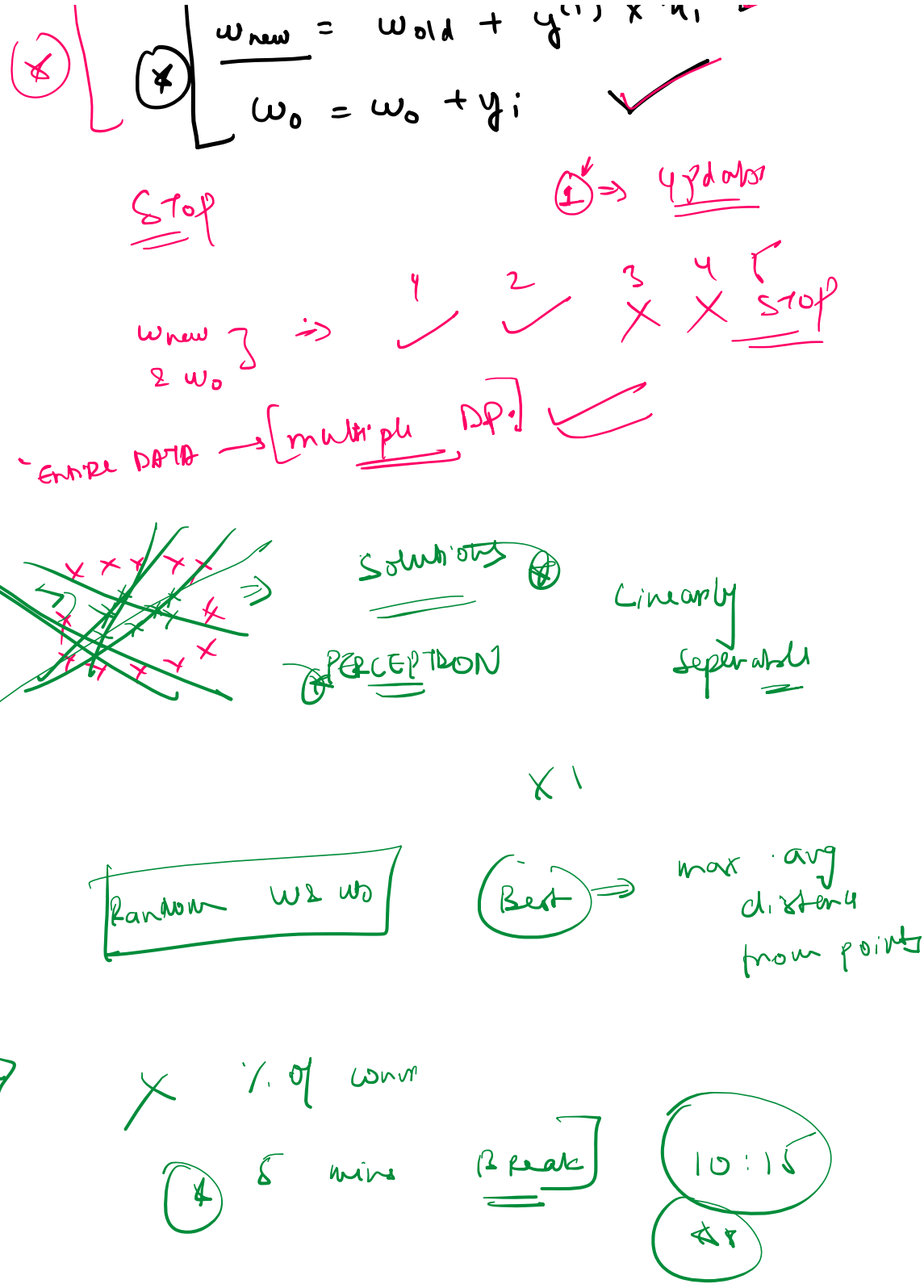
a lot of misclassified points



② iterate until convergence (till you have atleast 1 misclassified point) ⇒

update w & w_0 using

$w_{new} = w_{old} + y^{(i)} \times x_i$



If vector a=[3,4] and vector b=[5,0], what is the cos inverse of the angle between a and b?

4 options

Active Duration(Most preferred: 30 seconds)

Active Duration(Most preferred: 30 seconds)

$\vec{a} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$

$\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{\|a\| \|b\|} =$

$[3, 4] \begin{bmatrix} 5 \\ 0 \end{bmatrix}$

$= \dfrac{3 \times 5 + 4 \times 0}{\sqrt{3^2+4^2}\ \sqrt{5^2+0^2}}$

$= \dfrac{15}{\sqrt{25}\ \sqrt{25}} = \dfrac{15}{5 \times 5}$

$\cos\theta = 3/5 \Rightarrow \theta = \cos^{-1}\left(\frac{3}{5}\right)$

(4)